

35—BAR SCALES (continued)

BAR SCALE CALCULATIONS — MILES (1 mile = 63,360 inches)—continued							
FRACTIONAL SCALE	SCALE TO MAP REPRESENTATION		TO FIND MILES PER INCH (x in ratio)	MILES PER INCH	TOTAL MILES ON SCALE	TO FIND TOTAL SCALE LENGTH IN INCHES (y in ratio)	TOTAL SCALE LENGTH (INCHES)
	Scale Unit	represents Map Unit	Use ratio below or $\frac{\text{SCALE}}{63\,360}$			Use ratio below or $\frac{\text{Miles on scale}}{\text{Miles per inch}}$	
1:250 000	1 inch	250 000 in	$\frac{63\,360}{1} = \frac{250\,000}{x}$	3.945707	25	$\frac{3.945707}{1} = \frac{25}{y}$	6.336
1:400 000	1 inch	400 000 in	$\frac{63\,360}{1} = \frac{400\,000}{x}$	6.3131313	40	$\frac{6.3131313}{1} = \frac{40}{y}$	6.336
1:500 000	1 inch	500 000 in	$\frac{63\,360}{1} = \frac{500\,000}{x}$	7.8914141	50	$\frac{7.8914141}{1} = \frac{50}{y}$	6.336
1:750 000	1 inch	750 000 in	$\frac{63\,360}{1} = \frac{750\,000}{x}$	11.837121	60	$\frac{11.837121}{1} = \frac{60}{y}$	5.068
1:1 000 000	1 inch	1 000 000 in	$\frac{63\,360}{1} = \frac{1\,000\,000}{x}$	15.782828	125	$\frac{15.782828}{1} = \frac{125}{y}$	7.920
1:2 000 000	1 inch	2 000 000 in	$\frac{63\,360}{1} = \frac{2\,000\,000}{x}$	31.565656	250	$\frac{31.565656}{1} = \frac{250}{y}$	7.920
1:2 500 000	1 inch	2 500 000 in	$\frac{63\,360}{1} = \frac{2\,500\,000}{x}$	39.45707	300	$\frac{39.45707}{1} = \frac{300}{y}$	7.603
1:5 000 000	1 inch	5 000 000 in	$\frac{63\,360}{1} = \frac{5\,000\,000}{x}$	78.914141	600	$\frac{78.914141}{1} = \frac{600}{y}$	7.603
1:7 500 000	1 inch	7 500 000 in	$\frac{63\,360}{1} = \frac{7\,500\,000}{x}$	118.37121	600	$\frac{118.37121}{1} = \frac{600}{y}$	5.068
1:10 000 000	1 inch	10 000 000 in	$\frac{63\,360}{1} = \frac{10\,000\,000}{x}$	157.82828	1100	$\frac{157.82828}{1} = \frac{1100}{y}$	6.969
To find miles per inch on 1: 250 000 map . . . 63,360 inches = 1 mile Show in ratio as ... $\frac{63\,360}{1} \frac{\text{inches}}{\text{miles}}$ Let SCALE (250 000) be in inches Fractional scale says 1 inch represents 250,000 in Let x be miles that 1 inch represents on map Show in ratio as ... $\frac{250\,000}{x} \frac{\text{inches}}{\text{miles}}$ Solution . . . $\frac{63\,360}{1} = \frac{250\,000}{x}$ $\frac{63\,360}{63\,360} x = \frac{250\,000}{63\,360}$ $x = \frac{250\,000}{63\,360}$ (SCALE) $x = 3.945707$ $63\,360 \cdot x = 250\,000 \cdot 1$							